1. Finding Mean Using Python

```
data = [10, 20, 30, 40, 50]
mean_val = sum(data) / len(data)
print("Mean:", mean_val)
```

Output:

Mean: 30.0

2. Finding Median Using Python

```
data = [40, 10, 30, 50, 20]
data.sort()
n = len(data)
if n % 2 == 0:
    median_val = (data[n//2-1] + data[n//2]) / 2
else:
    median_val = data[n//2]
print("Median:", median_val)
```

Output:

Median: 30

3. Finding Mode Using Python

```
from collections import Counter
data = [1,2,2,3,4,2,5]
count = Counter(data)
mode_val = count.most_common(1)[0][0]
print("Mode:", mode_val)
```

Output:

Mode: 2

4. Finding Range Using Python

```
data=[15,8,22,42,10]
range_val=max(data)-min(data)
print("Range:", range_val)
```

Output:

Range: 34

5. Finding Variance Using Python

```
data=[2,4,4,4,5,5,7,9]
mean_val=sum(data)/len(data)
variance_val=sum((x-mean_val)**2 for x in data)/len(data)
print("Variance:", variance_val)
```

Output:

Variance: 4.0

6. Finding Standard Deviation

```
data=[2,4,4,4,5,5,7,9]
mean_val=sum(data)/len(data)
variance_val=sum((x-mean_val)**2 for x in data)/len(data)
std_dev=variance_val**0.5
print("Standard Deviation:", std_dev)
```

Output:

Standard Deviation: 2.0

7. Using statistics Module

```
import statistics
data=[12,15,15,18,22,25]
print("Mean:", statistics.mean(data))
print("Median:", statistics.median(data))
print("Mode:", statistics.mode(data))
print("Variance:", statistics.variance(data))
print("Standard Deviation:", statistics.stdev(data))
```

Output:
Mean: 17.833333333333332
Median: 16.5
Mode: 15
Variance: 24.166666666666668
Standard Deviation: 4.915992779

8. Using NumPy

```
import numpy as np
data=np.array([10,20,20,30,40])
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Variance (Sample):", np.var(data,ddof=1))
print("Standard Deviation:", np.std(data,ddof=1))
print("Range:", np.ptp(data))
```

Output:
Mean: 24.0
Median: 20.0
Variance (Sample): 130.0
Standard Deviation: 11.40175425099138
Range: 30

9. Using SciPy for Mode

```
from scipy import stats
data=[5,2,5,3,5,4]
mode_result=stats.mode(data, keepdims=True)
print("Mode:", mode_result.mode[0])
print("Count:", mode_result.count[0])
```

Output:
Mode: 5
Count: 3

10. Using Pandas

```
import pandas as pd
df=pd.DataFrame({"values":[10,20,30,40,50,50]})
print("Mean:", df["values"].mean())
print("Median:", df["values"].median())
print("Mode:", df["values"].mode().tolist())
print("Variance:", df["values"].var())
print("Std Dev:", df["values"].std())
print("Range:", df["values"].max()-df["values"].min())
```

Output:

```
Mean: 33.333333333333336
Median: 35.0
Mode: [50]
Variance: 266.6666666666667
Std Dev: 16.32993161855452
Range: 40
```